MZUZU UNIVERSITY

FACULTY OF SCIENCE, TECHNOLOGY AND INNOVATION

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

BACHELOR OF SCIENCE IN INFORMATION AND COMMUNICATION TECHNOLOGY

BICT 4701: SOFTWARE ENGINEERING

END OF SEMESTER EXAMINATION

DATE: AUGUST 2022 **TIME ALLOWED: 3 HOURS**

INSTRUCTIONS

i. There are 12 questions in this paper. **ANSWER ALL.**

ii. There are **TWO** pages. **PLEASE CHECK**

iii. Marks are indicated for each question.

1. Define the following terms:
    a. Software Engineering. **[1 Mark]**
    b. Adaptability. **[1 Mark]**
    c. Legacy Software. **[1 Mark]**
    d. Software Crisis. **[1 Mark]**
    e. Software Myths **[1 Mark]**
2. Explain **three** components that encompass any software product. **[3 Marks]**
3. Differentiate between the following:
    a. Scrum and Extreme Programming. **[2 Marks]**
    b. Prototype model and Incremental model. **[2 Marks]**
    c. Prescriptive model and agile model. **[2 Marks]**
4. Describe **three** parameters used to assess or validate a software process model. **[6 marks]**
5. Explain **two** roles of refactoring in a software engineering process. **[4 Marks]**
6. Explain **two** advantages that come with writing tests before writing the code. **[4 Marks]**
7. Write test cases for the following system features:
    a. Dose checking in a mental health management system. **[3 Marks]**
    b. User authentication in any information system. **[3 Marks]**
    c. User registration in a semester registration system. **[3 Marks]**
8. How does each of the XP values affect the software development process and its outcome? **[10 Marks]**
9. Explain any **three** examples of process model flow. **[6 Marks]**
10. Describe Hooker's general principles of software engineering practice and relate them to the software development process. **[14 Marks]**
11. Describe the process of requirements engineering. **[8 Marks]**
12. Read the user story for Library Management System below and use it to answer questions that follow:

    Mzuzu University wishes to develop a library management system. The system will allow students and staff to borrow and return books from the library. The system will also allow the users to reserve and renew books. The system will allow users to register online and search available books.

The system will issue a fine to users who will not return the books in time. The system will also allow the administrator to manage books and user accounts. The University has recommended you work on the project as a systems engineer.

a. Draw a use case diagram that can be used to elicit software requirements. **[10 Marks]**

b. Use "Reserve Book" functional requirement to draw a sequence diagram **[8 Marks]**

c. Draw a class diagram with each class showing the class name, attributes, and operations. **[12 Marks]**

END OF QUESTION PAPER